

DATA ANALYTICS CASE STUDY

**Bank Marketing Dataset
Data Analysis
Term Deposit Subscription**

Dataset: bank-full.csv | 45,211 records | Target: y
August, 2026

Executive Summary

This analysis evaluates customer and campaign factors that influence term deposit subscription in a banking direct-marketing context. The overall conversion rate is 11.70%, indicating that most campaign contacts do not result in subscription and that better targeting can materially improve commercial efficiency. The strongest business signals are previous campaign success, customer engagement during the call, contact channel, campaign timing, and selected occupational segments. Customers with a previous successful campaign outcome show a substantially higher likelihood of subscribing, while cellular contacts outperform other channels. The findings suggest that the bank should prioritize customers with positive prior interactions, optimize outreach through stronger channels, and focus campaign resources on segments with higher observed response potential. By applying these insights to targeting, scheduling, and lead prioritization, marketing teams can reduce low-probability contacts, improve conversion efficiency, and support more profitable customer acquisition.

Stage 1 - Ask

Business task

A financial institution wants to improve the effectiveness of direct marketing campaigns by identifying the customer, financial, and campaign characteristics most strongly associated with term deposit subscription.

Primary business question

Which customer, financial, and campaign characteristics are most strongly associated with term deposit subscription?

Stakeholders and success metrics

- Stakeholders: Marketing Department, Sales Management, Investment Products Team, and Executive Leadership.
- Success metrics: conversion rate, campaign efficiency, segmentation accuracy, and predictive performance for the subscription outcome.

Stage 2 - Prepare

Dataset description

The dataset is related to direct marketing campaigns of a Portuguese banking institution. The campaign objective is to predict whether the client subscribed to a term deposit. The source documentation states that bank-full.csv contains all examples, ordered by date from May 2008 to November 2010, with 45,211 instances and 16 input attributes plus the output variable y.

45,211	11.70%	5,289	537.3s
Total records	Conversion rate	Subscribers	Avg duration, subscribers

Data area	Variables used
Customer demographics	age, job, marital, education
Financial relationship	default, balance, housing, loan
Campaign contact	contact, day, month, duration
Campaign history	campaign, pdays, previous, poutcome
Outcome	y: subscribed to a term deposit, yes/no

Credibility note

The dataset is public and cited in Moro, Laureano and Cortez, 2011. The source documentation states there are no missing attribute values, although values such as unknown appear as valid categories and should be handled explicitly during analysis.

Stage 3 - Process

Cleaning and validation steps

- Loaded the semicolon-delimited CSV file and validated column structure.
- Checked the target variable y and confirmed it contains two classes: yes and no.
- Reviewed numeric fields such as age, balance, duration, campaign, pdays, and previous.
- Reviewed categorical fields and retained unknown as a category rather than deleting records.
- Confirmed the analysis preserved all 45,211 records from bank-full.csv.
- Separated interpretation of raw fields from engineered business features, such as previous-contact flags and contact-frequency bands.

Important processing decisions

Issue	Treatment
unknown categories	Kept as valid categorical values to avoid removing observations.
pdays = -1	Interpreted as not previously contacted, based on the source documentation.
duration	Used for explanatory analysis; flagged as a potential leakage variable for pre-call prediction.
class imbalance	Reported directly and recommended imbalance-aware evaluation metrics.

Stage 4 - Analyze

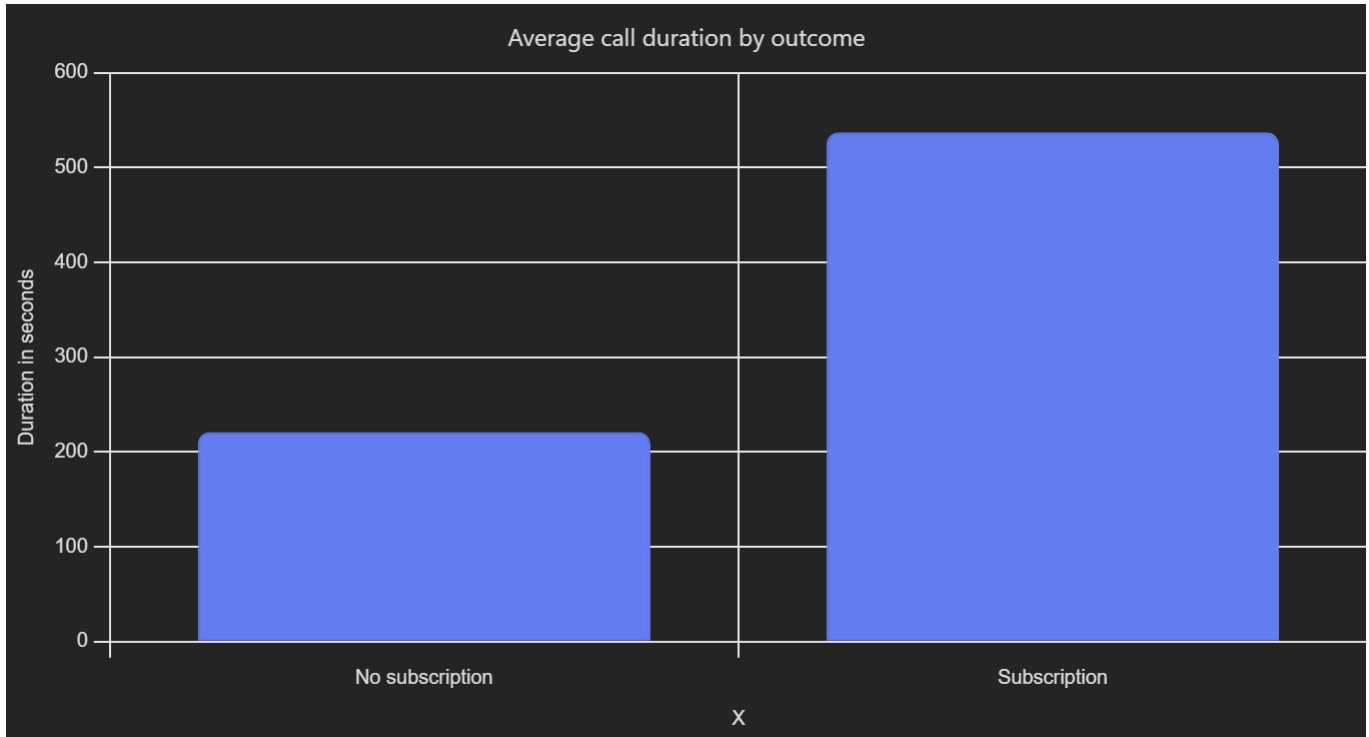
Target distribution

Outcome	Records	Share
No subscription	39,922	88.30%
Subscription	5,289	11.70%

The dataset is imbalanced toward non-subscription. This means that accuracy alone would not be sufficient for predictive modeling; precision, recall, F1-score, ROC-AUC, and PR-AUC are more appropriate evaluation metrics.

Finding 1: Call duration is strongly associated with conversion

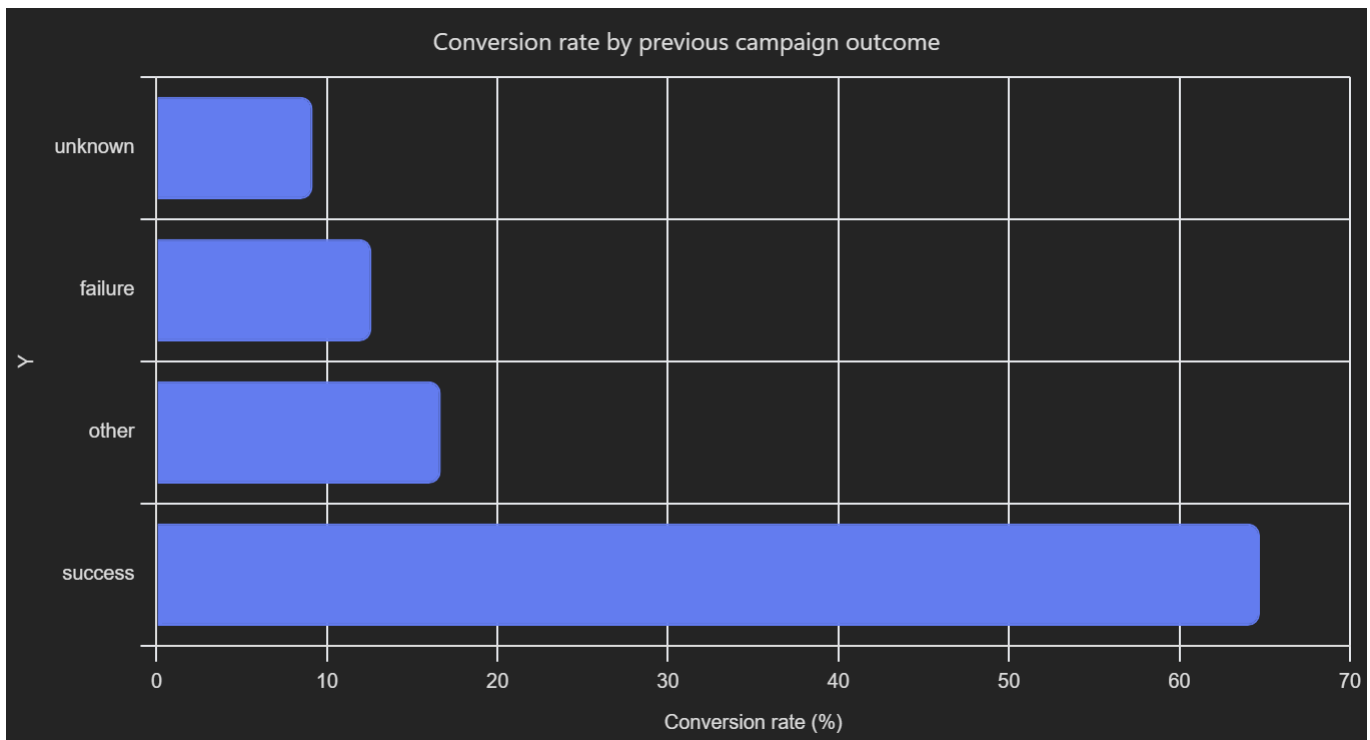
Customers who subscribed had an average call duration of 537.3 seconds, compared with 221.2 seconds for customers who did not subscribe. This suggests that engagement during the call is strongly associated with conversion. However, duration should be used carefully because it is only known after the call occurs and may create leakage if the goal is to predict before the call.



Finding 2: Previous campaign outcome is a strong signal

Customers with a previous campaign outcome of success show a much higher conversion rate than other previous-outcome categories. This indicates that campaign history is valuable for lead prioritization.

Previous outcome	Conversion rate
success	64.73%
other	16.68%
failure	12.61%
unknown	9.16%



Finding 3: Contact channel matters

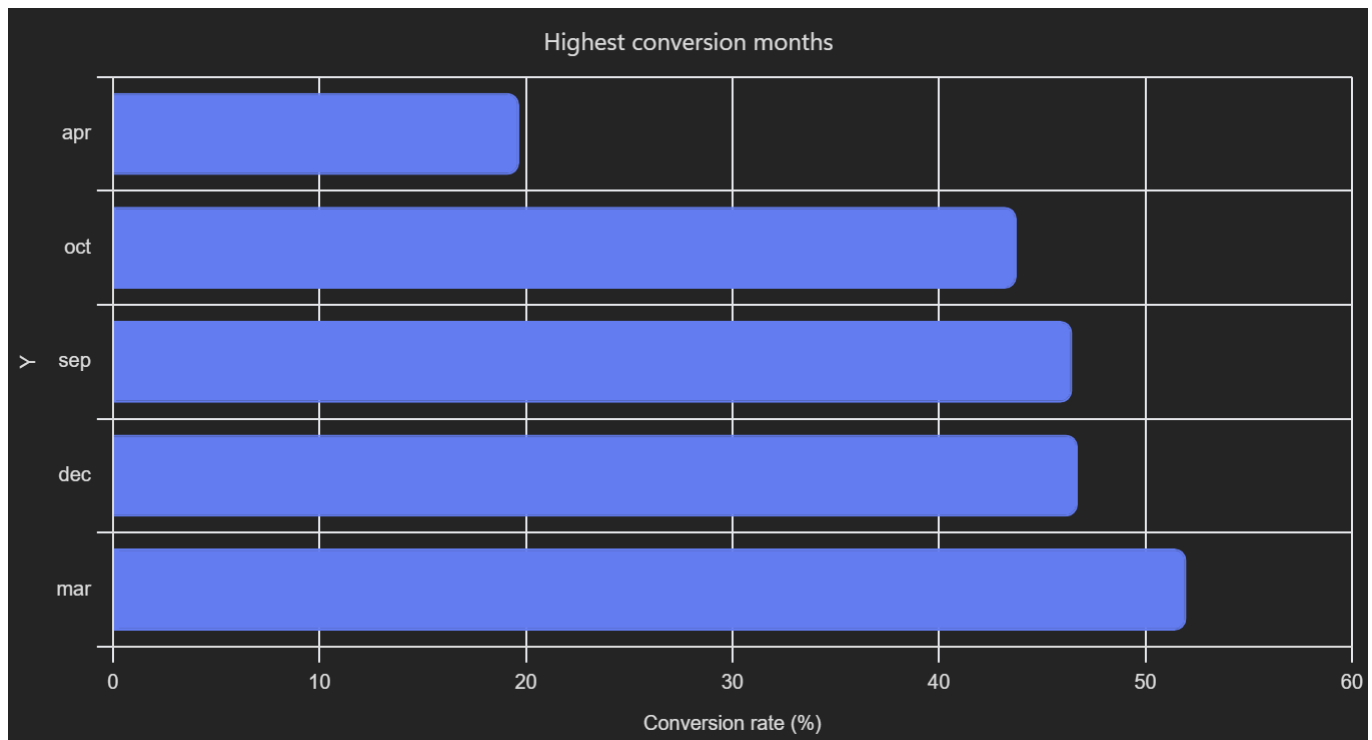
Contact type	Conversion rate
cellular	14.92%
telephone	13.42%
unknown	4.07%

Cellular contact has the highest observed conversion rate among the contact categories, while records with unknown contact type have the lowest conversion rate.

Finding 4: Conversion varies strongly by month

Conversion rates vary materially across months. March, December, September, and October show the highest observed rates in the analysis. This suggests that campaign timing should be treated as a strategic decision.

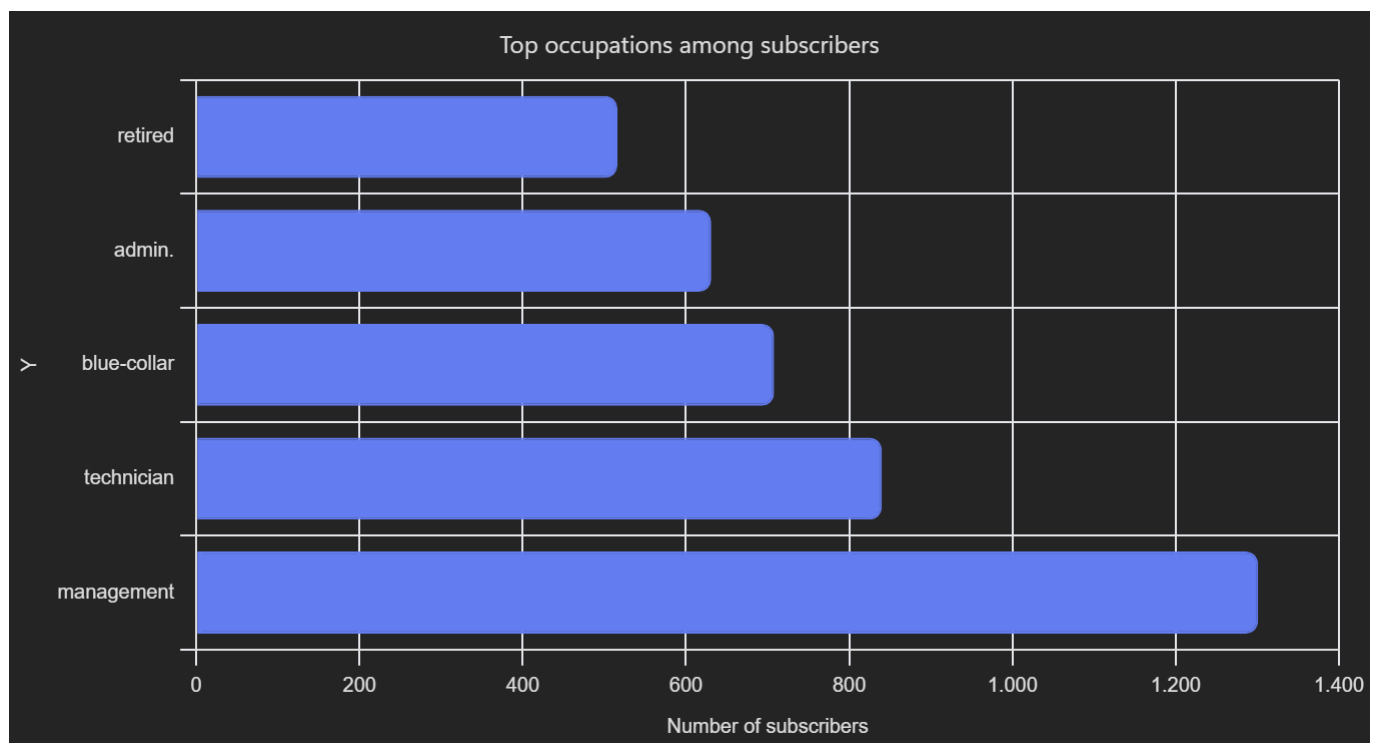
Month	Conversion rate
mar	51.99%
dec	46.73%
sep	46.46%
oct	43.77%
apr	19.68%



Finding 5: Occupation helps define target segments

The highest absolute number of subscribers appears among management, technician, blue-collar, admin., and retired customers. This does not necessarily mean these occupations have the highest conversion rates, but it identifies large subscriber pools that deserve segmentation analysis.

Occupation	Subscribers
management	1301
technician	840
blue-collar	708
admin.	631
retired	516



Finding 6: Age alone is not a strong differentiator

The average age of subscribers is 41.7, while the average age of non-subscribers is 40.8. Since the difference is small, age alone is less useful than behavior and campaign-history variables.

Stage 5 - Share & Act

Key insights

- Previous campaign success is one of the strongest indicators of future conversion.
- Longer call duration is strongly associated with subscription, although it should be handled carefully in predictive modeling.
- Cellular contact outperforms telephone and unknown contact categories.

- Conversion varies significantly by month, indicating seasonality and timing effects.
- Occupation helps identify large subscriber pools, especially management, technician, blue-collar, admin., and retired customers.

Business recommendations

- Prioritize customers with previous campaign success.
- Use cellular contact channels where possible.
- Develop tailored campaigns for management professionals, technicians, blue-collar customers, admin. profiles, and retirees.
- Plan campaigns around historically stronger months and test whether seasonality persists in future data.
- Build a predictive model using Logistic Regression as a baseline and Random Forest or Gradient Boosting for comparison.
- Use stratified validation and evaluate precision, recall, F1-score, ROC-AUC, and PR-AUC due to class imbalance.

Final conclusion

The analysis confirms that term deposit subscription can be partially explained by campaign history, engagement, contact channel, campaign timing, and customer occupation. The most actionable finding is that historical response and customer engagement should guide campaign prioritization. A data-driven campaign strategy can reduce low-probability contacts and increase the efficiency of marketing resources.

References and source notes

Dataset documentation: bank-names.txt. The source describes the Bank Marketing dataset, its attributes, citation request, and the target variable y. Citation requested by the source: Moro, Laureano and Cortez, 2011, Using Data Mining for Bank Direct Marketing: An Application of the CRISP-DM Methodology.

Analysis file used: bank-full.csv. Counts, conversion rates, and charts in this report were computed from that file during the prior analysis step.